

PROJECT: BEE-1012

sub-system: EXP

Version: 1.0.0



Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH		Sheet title: 01_COVER PAGE.SchDoc	Sheet 1 of 33
Checked by: BAO BUI		Document No: 1	Size: A3
Approved by: VU PHAM		Revision: 1.0.0	Date: 1/23/2026

1	2	3	4	5	6
A					
B					
C					
D					
1	2	3	4	5	6

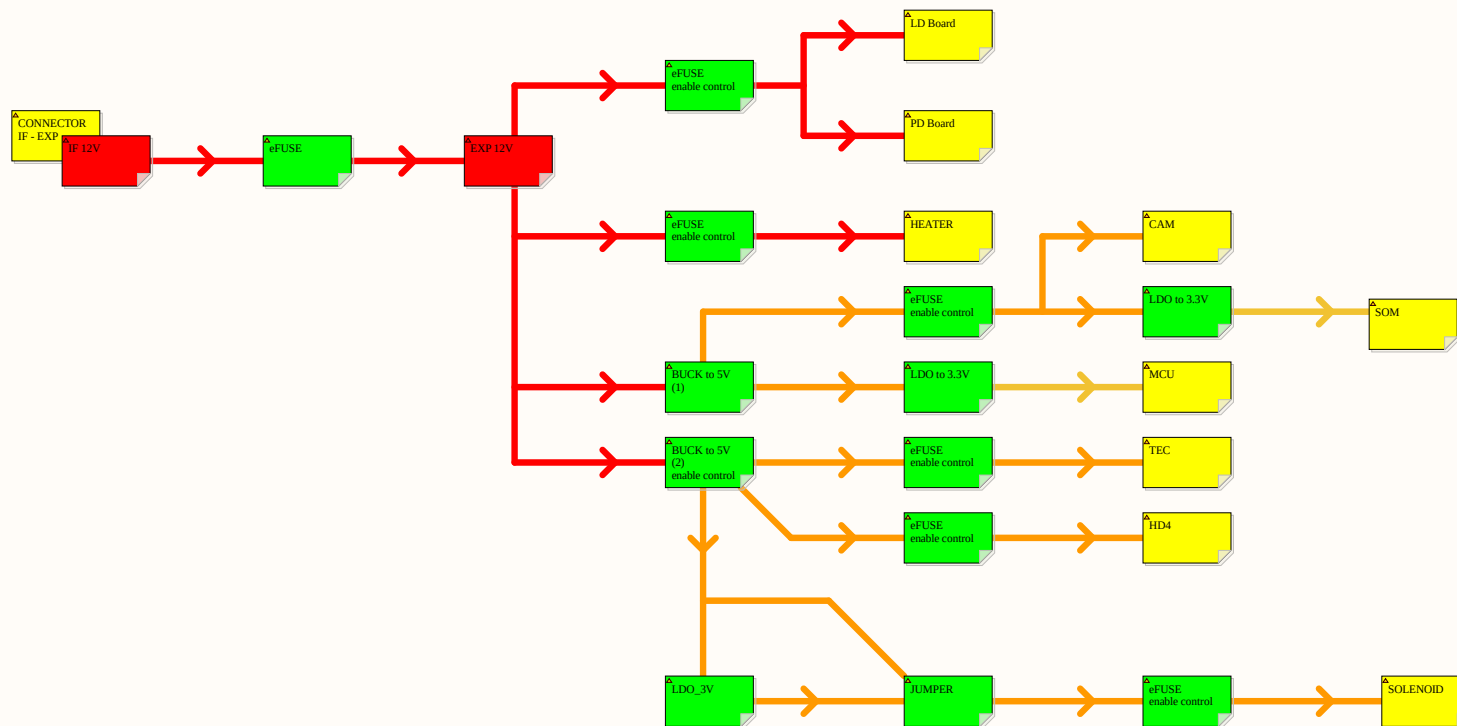
		
Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: 02_BLOCK DIAGRAM.Sch	Sheet 2 of 33
Checked by: BAO BUI	Document No: 2	Size: A3
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026

INDEX

Page	Title	Page	Title	Page	Title
1	COVER PAGE	11		21	
2	BLOCK DIAGRAM	12		22	
3	INDEX	13		23	
4	PI HAT	14		24	
5		15		25	
6		16		26	
7		17		27	
8		18		28	
9		19		29	
10		20		30	

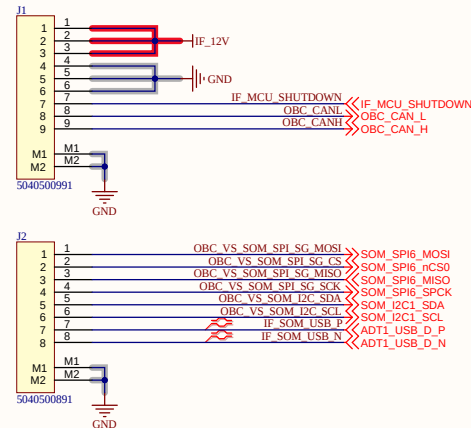


Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 03_INDEX.SchDoc	Sheet 3 of 33	
Checked by: BAO BUI	Document No: 3	Size: A3	
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026	

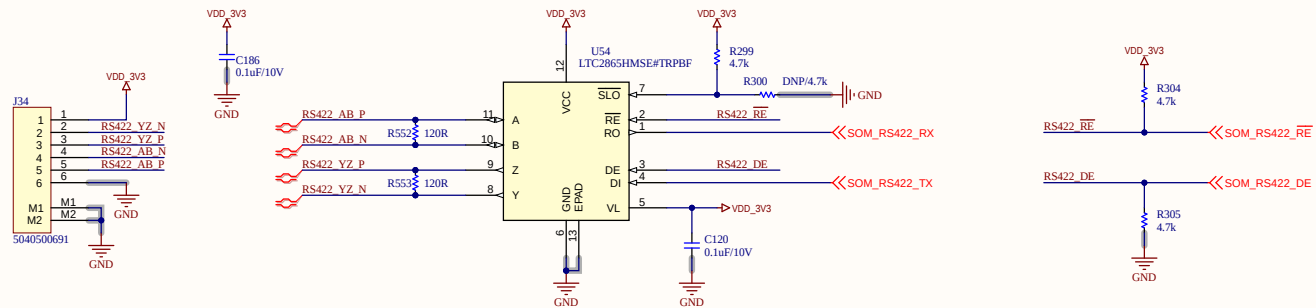


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 04_POWER_TREE.SchDoc	Sheet 4 of 33
Checked by: BAO BUI	Document No: 4	Size: A3
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026

form IF (OBC Board)

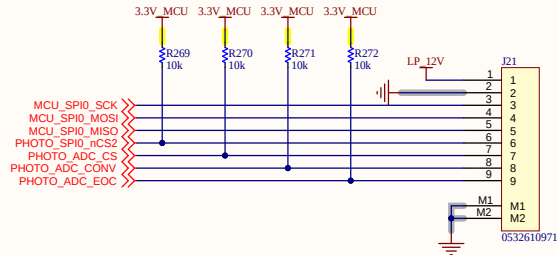


RS422

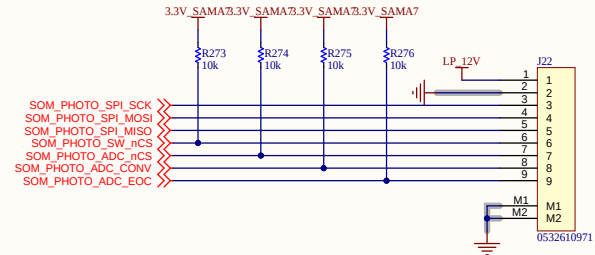


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: 05_CONNECTOR_IN.SchDoc
Checked by: BAO BUI	Document No: 5
Approved by: VU PHAM	Revision: 1.0.0
	Size: A3
	Date: 1/23/2026

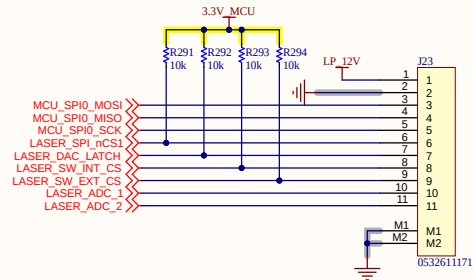
for PHOTO board



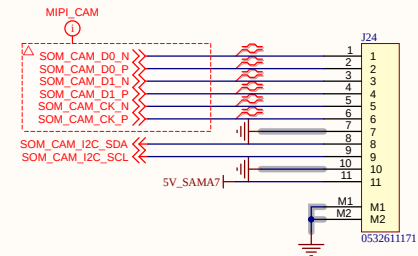
SOM to PHOTO



for LASER board



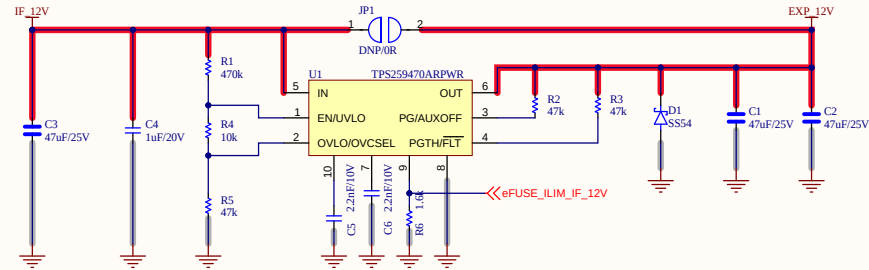
for CAMERA



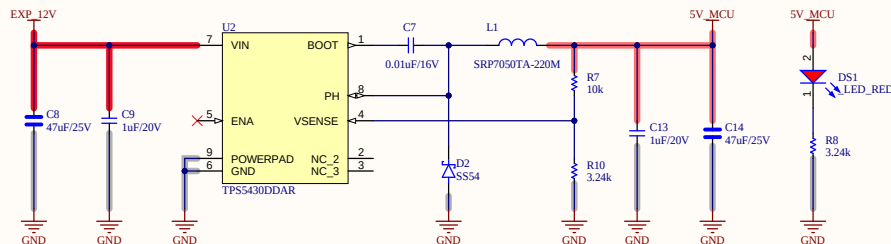
Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: 05_CONNECTOR_OUT.Schema 6 of 33
Checked by: BAO BUI	Document No: 6
Approved by: VU PHAM	Revision: 1.0.0
	Date: 1/23/2026

EFUSE 12V

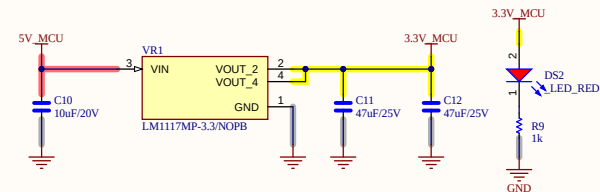
Undervoltage Protector: NONE
 Overvoltage Protector: 1.2*(47+470)/47=13.2V
 Slew Rate: 2000/2200=0.91V/ms
 I-LIM: 3334/1600=2.08A



BUCK for MCU, SOM

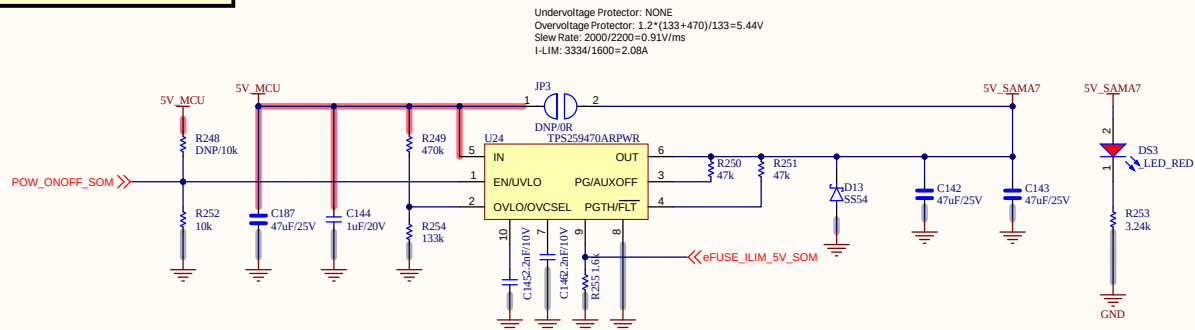


LDO for MCU

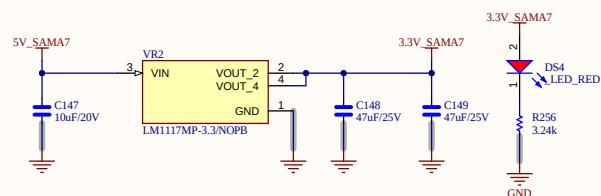


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 06_POWER_1.SchDoc	Sheet 7 of 33
Checked by: BAO BUI	Document No: 7	Size: A3
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026

EFUSE for SOM

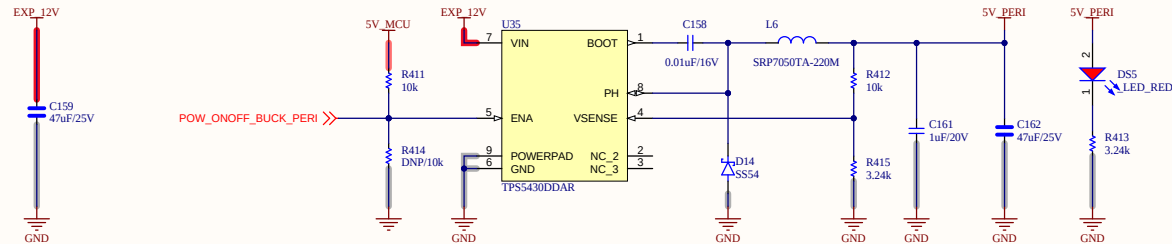


LD0 for SOM

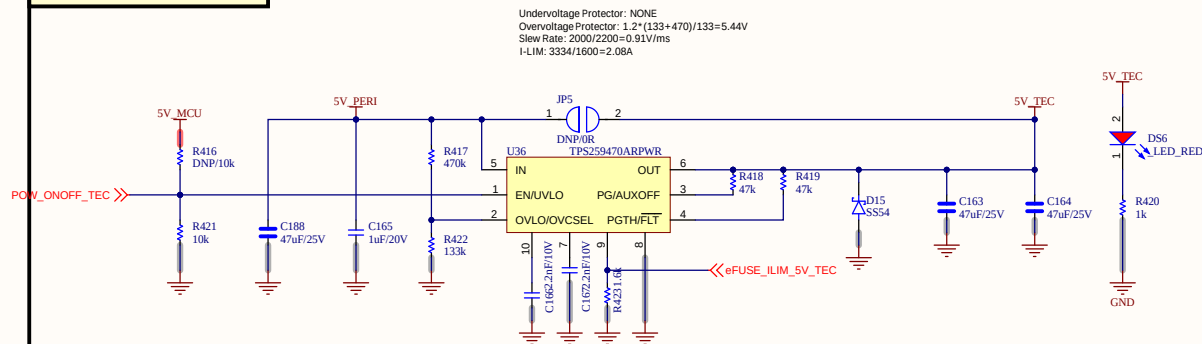


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER		
Design by: SANG HUYNH	Sheet title: 06_POWER_2.SchDoc	Sheet 8 of 33	
Checked by: BAO BUI	Document No: 8	Size: A3	
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026	

BUCK for PERIPHERAL

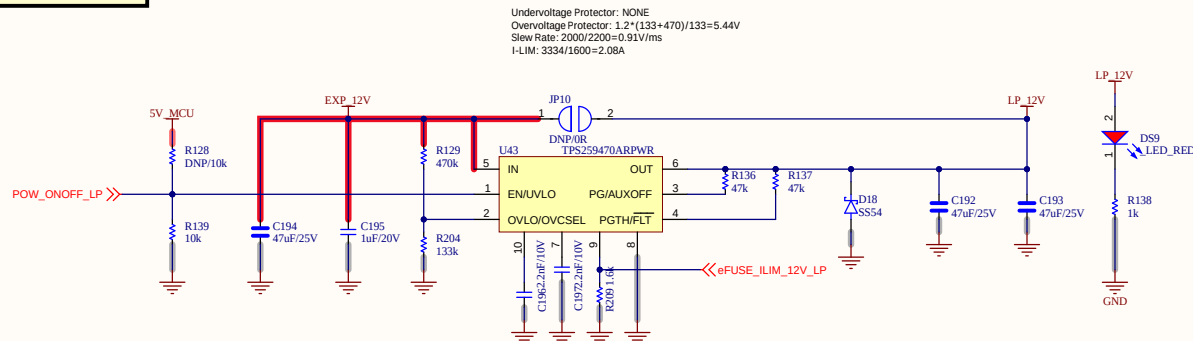


EFUSE for TEC

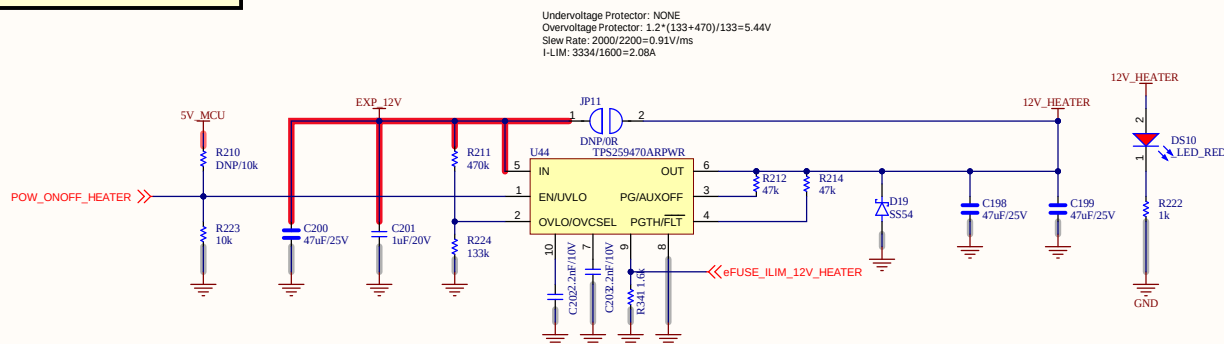


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 06_POWER_3.SchDoc	Sheet 9 of 33
Checked by: BAO BUI	Document No: 9	Size: A3
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026

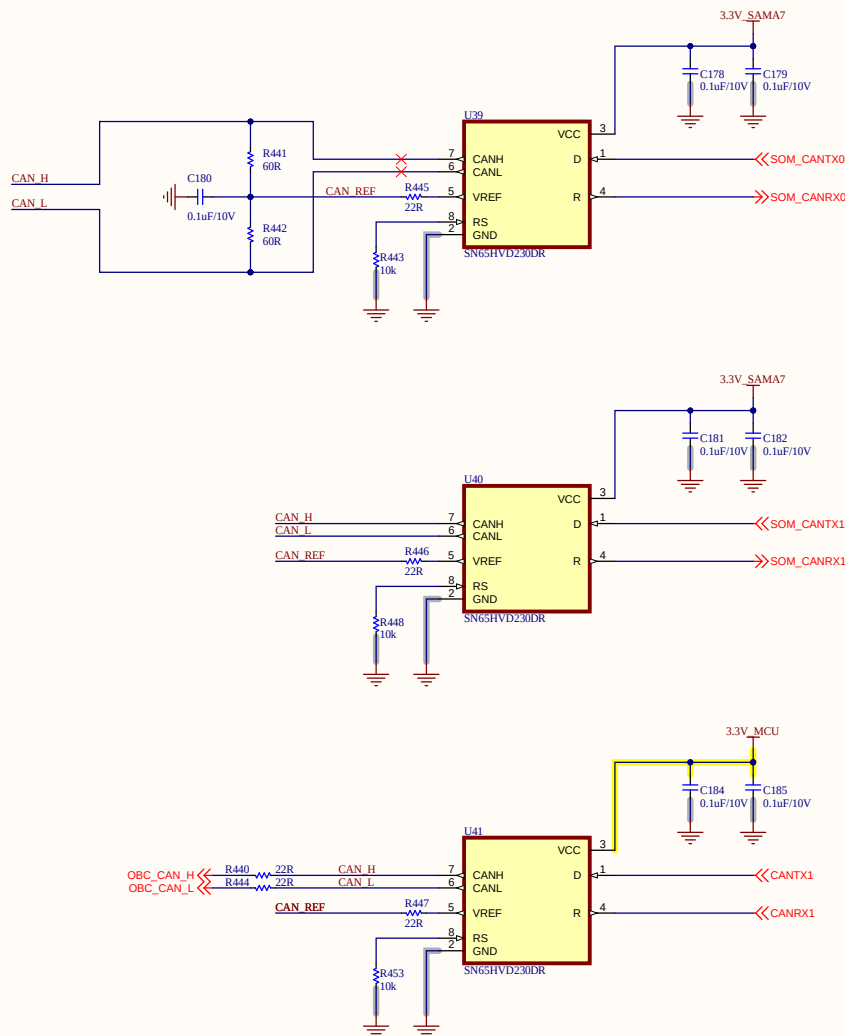
EFUSE for LP



EFUSE for HEATER

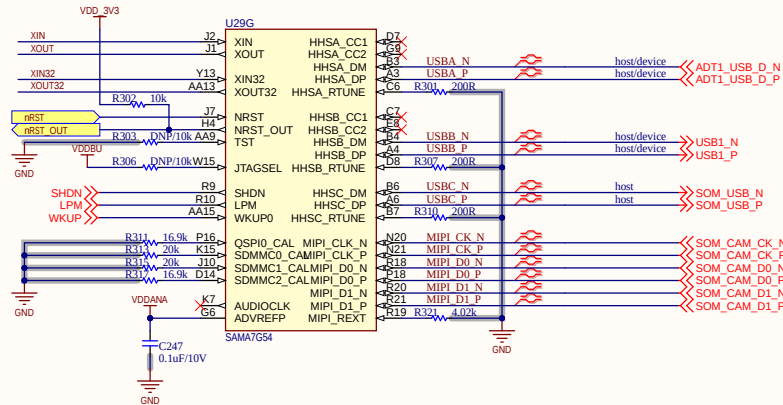


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER		
Design by: SANG HUYNH	Sheet title: 06_POWER_5.SchDoc	Sheet 11 of 33	
Checked by: BAO BUI	Document No: 11	Size: A3	
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026	

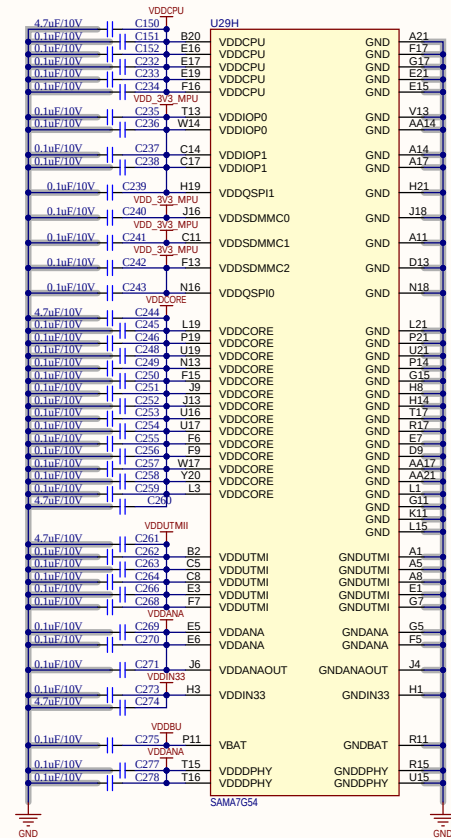


Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 07_CAN.SchDoc	Sheet 12 of 33	
Checked by: BAO BUI	Document No: 12	Size: A3	
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026	

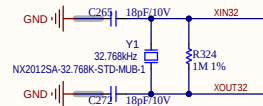
MPU configuration



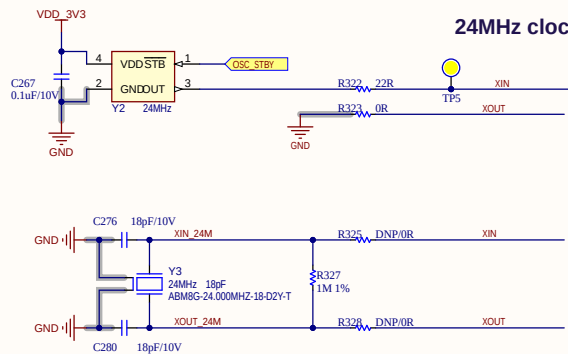
MPU decoupling



32.768KHz clock



24MHz clock



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 08_SAMA7G54-MPU.SchDoc Sheet 13 of 33

Checked by: BAO BUI

Document No: 13

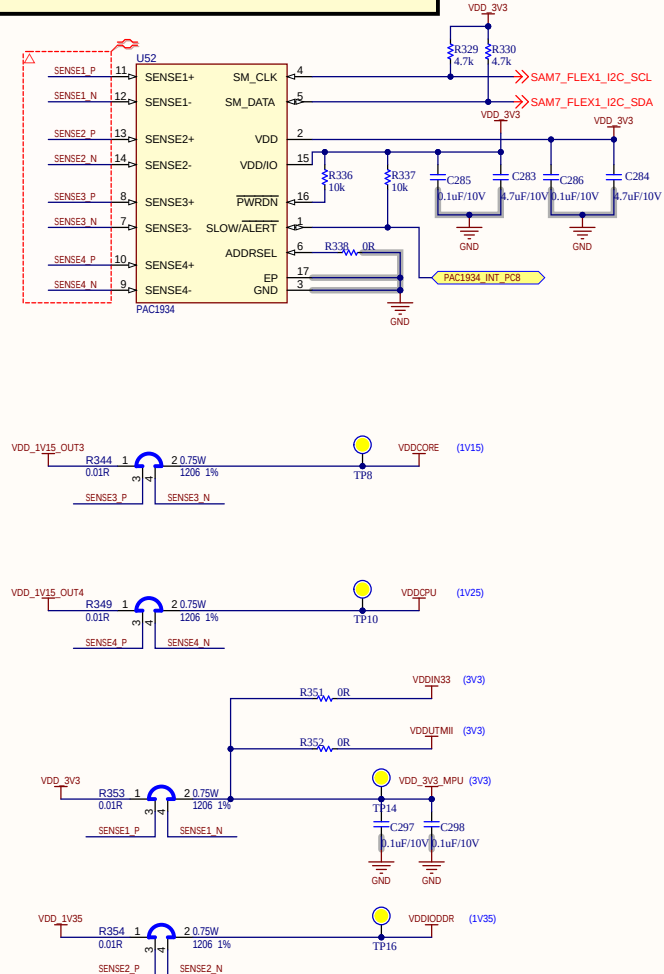
Size: A3

Approved by: VU PHAM

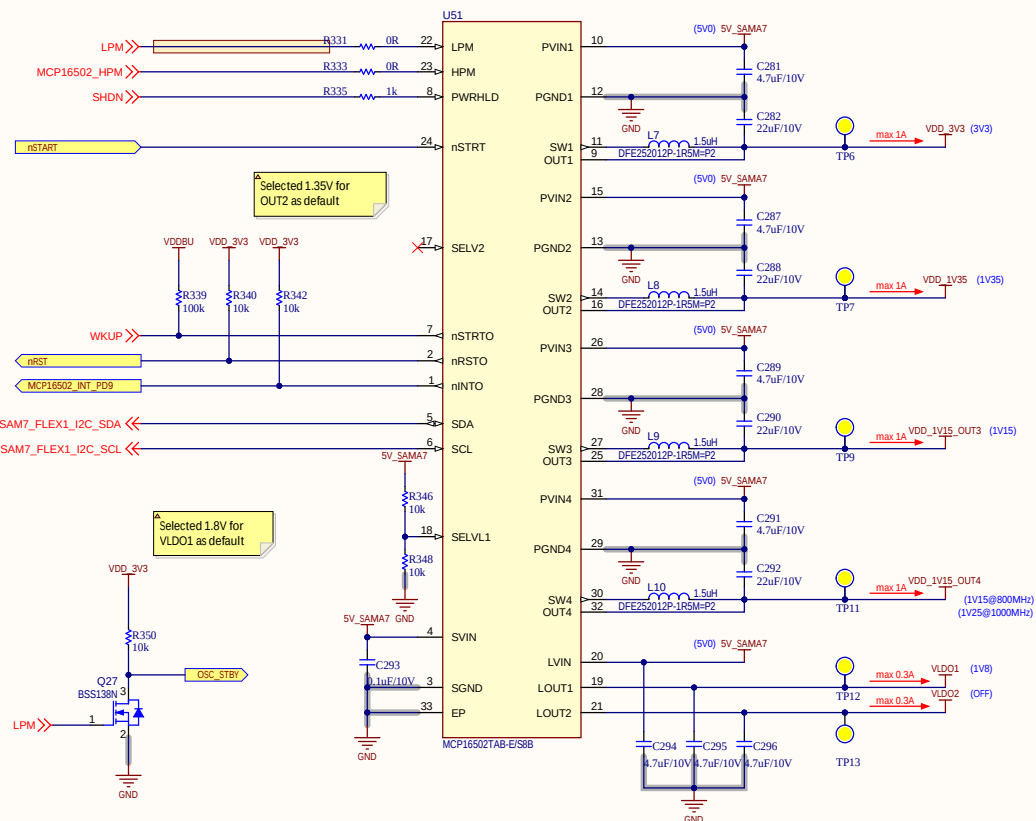
Revision: 1.0.0

Date: 1/23/2026

Power Measurement and Distribution

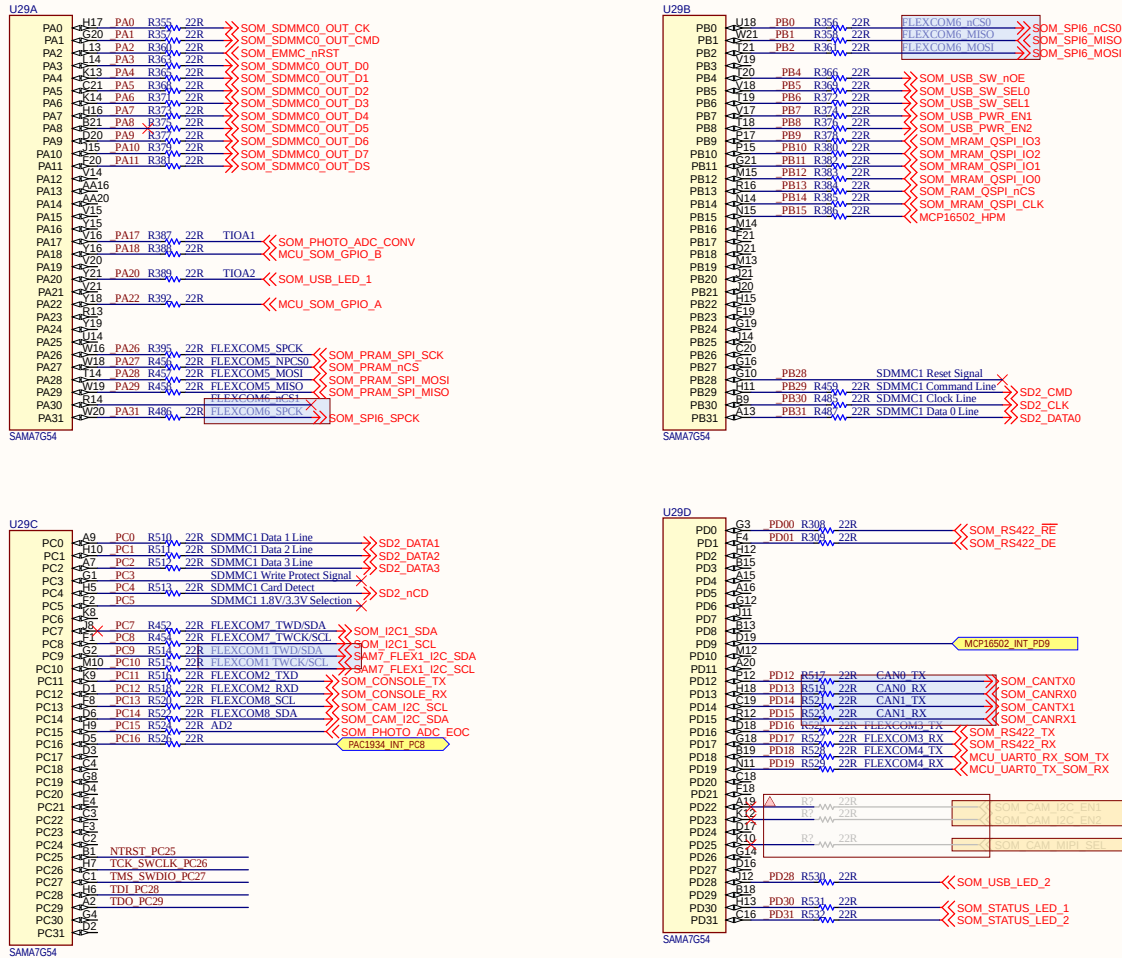


PMIC

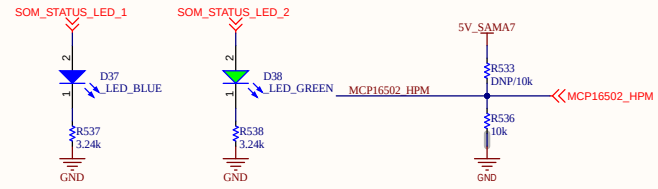


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: 08_SAMA7G54-PMIC.SchDoc
Checked by: BAO BUI	Document No: 14
Approved by: VU PHAM	Revision: 1.0.0
	Size: A3
	Date: 1/23/2026

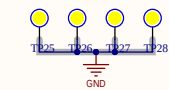
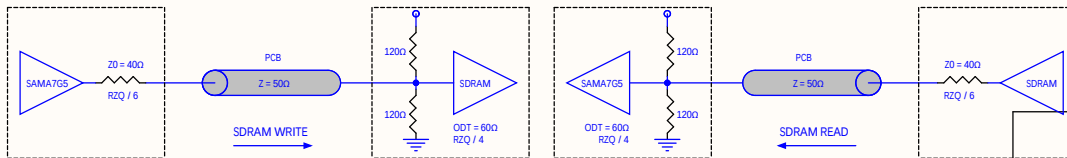
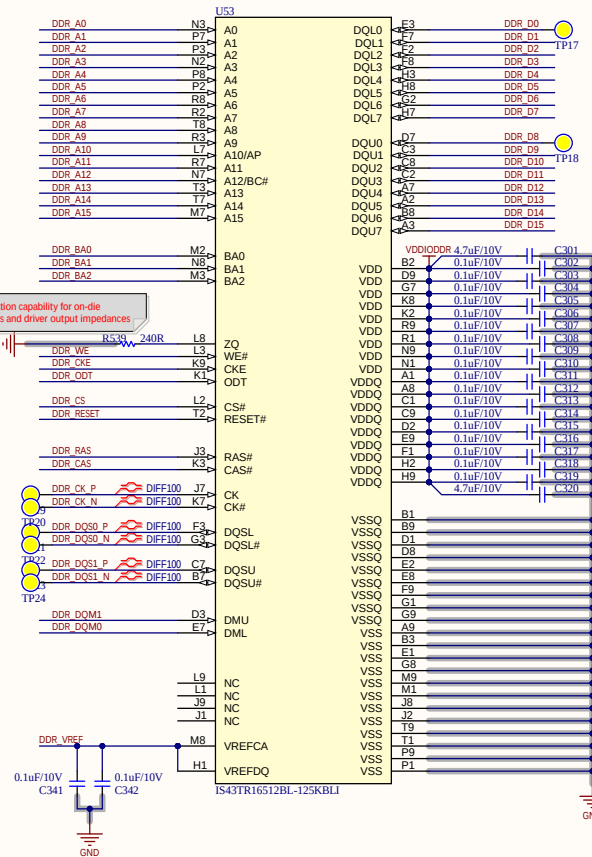
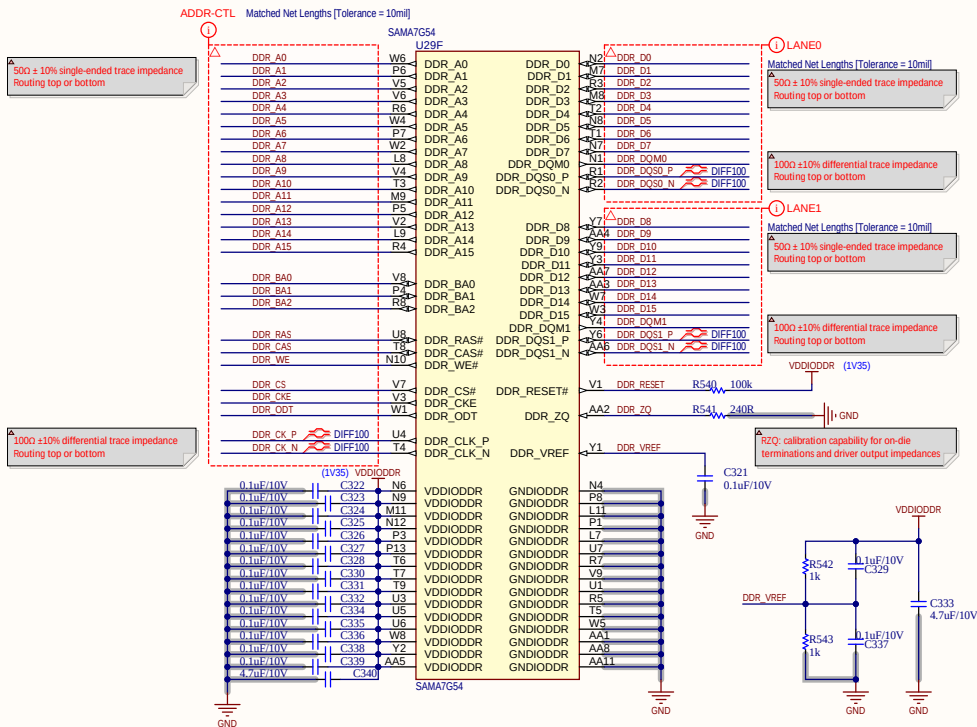
SAMA7G5 Bus Distribution



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER		
Design by: SANG HUYNH	Sheet title: 08_SAMA7G54-PIO.SchDoc		
Checked by: BAO BUI	Document No: 15	Size: A3	
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026	



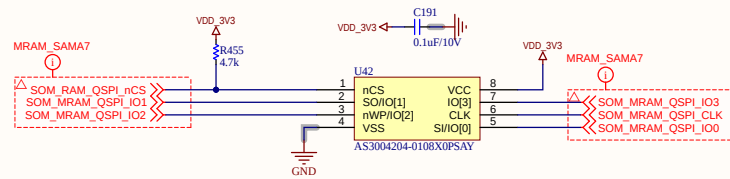
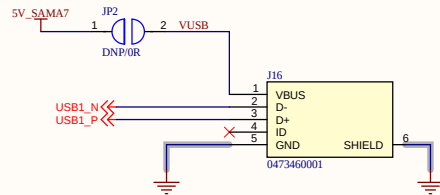
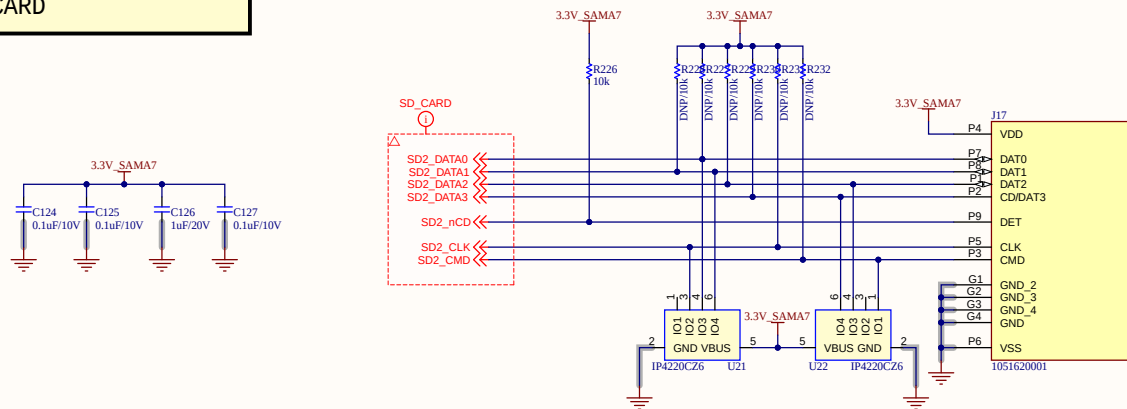
SAMA7G5 DDR3-1066 controller



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: 08_SAMA7G54-DDR3.Sch
Checked by: BAO BUI	Document No: 16
Approved by: VU PHAM	Revision: 1.0.0
	Size: A3
	Date: 1/23/2026

USB

The diagram illustrates the USB connection for the 0473460001 module. A 5V_SAMA7 source is connected to pin 1 of JP2. Pin 2 of JP2 is connected to VUSB. USB1_N and USB1_P are connected to pins 2 and 3 of J16. Pin 4 of J16 is connected to GND. Pin 5 of J16 is connected to GND. Pin 6 of J16 is connected to GND. The module is labeled 0473460001 and SHIELD.

[illegible]

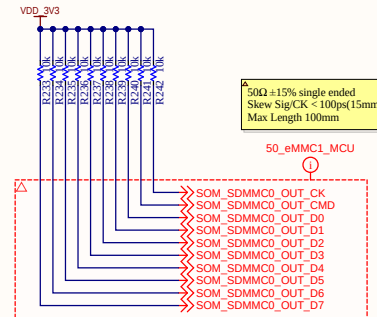
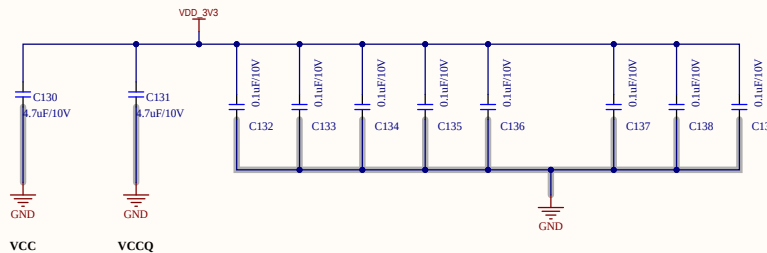
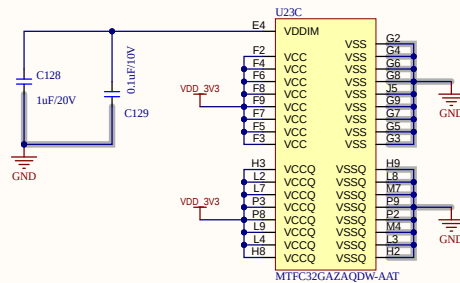
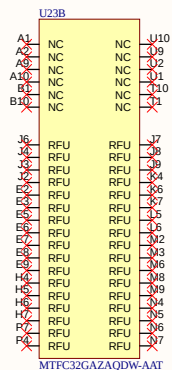
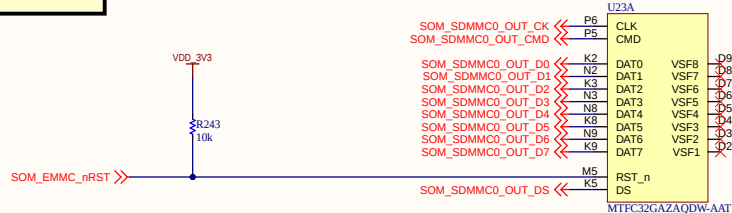
Sub-system: SATELLITE_EXP_CONTROLLER

Sheet title: 08_SOM_USB_CARD.Sch Sheet 17 of 33

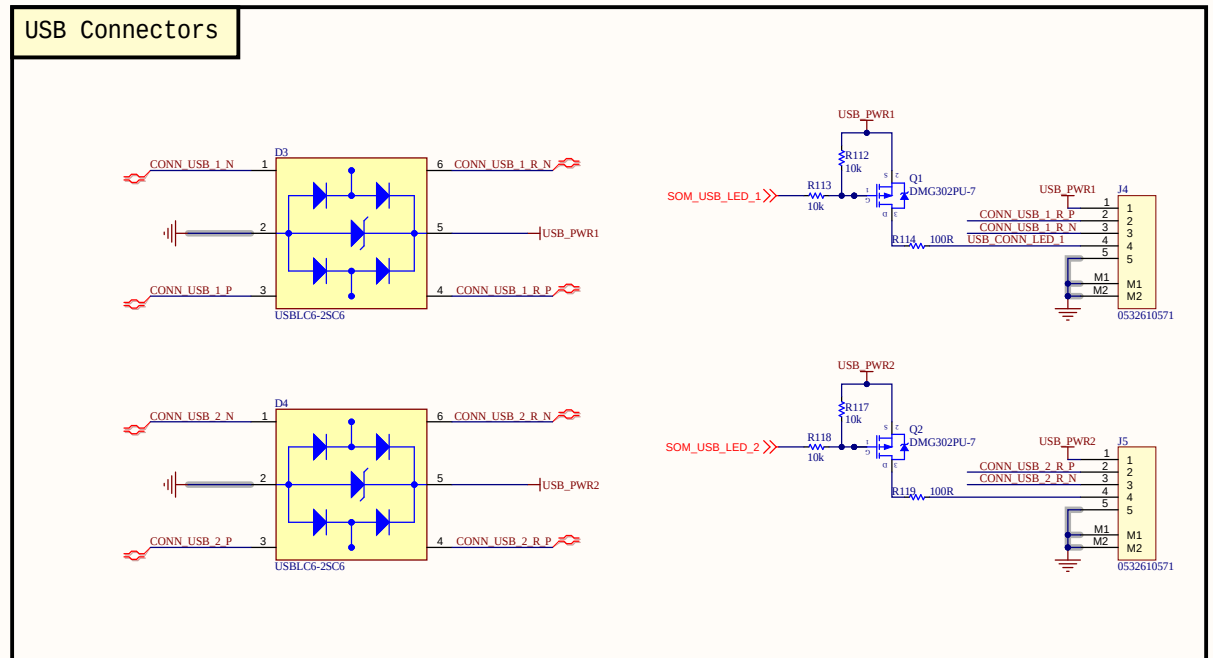
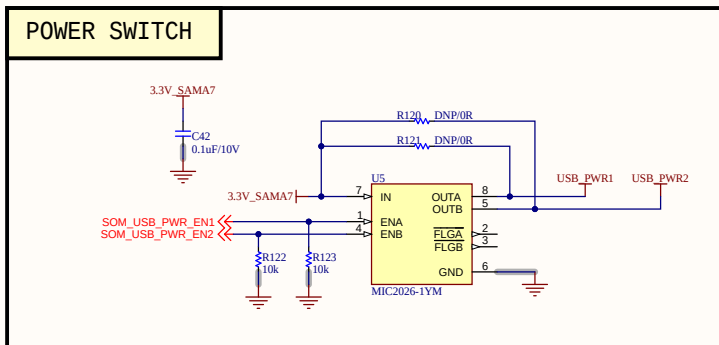
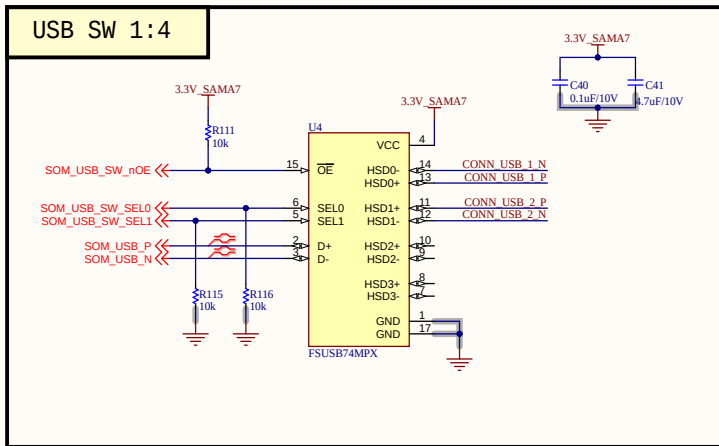
Size: A3

Date: 1/23/2026

eMMC - Buffer



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 09_eMMC.SchDoc	Sheet 18 of 33
Checked by: BAO BUI	Document No: 18	Size: A3
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026

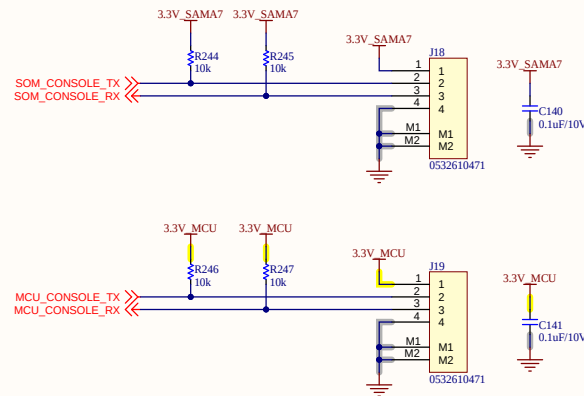


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 10_USB.SchDoc	Sheet 19 of 33
Checked by: BAO BUI	Document No: 19	Size: A3
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026



Date: 1/23/2026

CONSOLE



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 12_CONSOLE.SchDoc

Sheet 21 of 33

Checked by: BAO BUI

Document No: 21

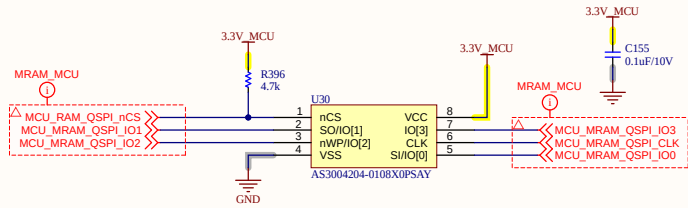
Size: A3

Approved by: VU PHAM

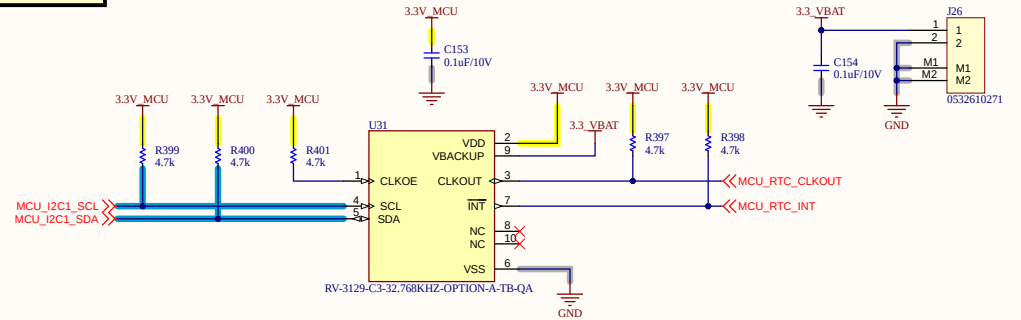
Revision: 1.0.0

Date: 1/23/2026

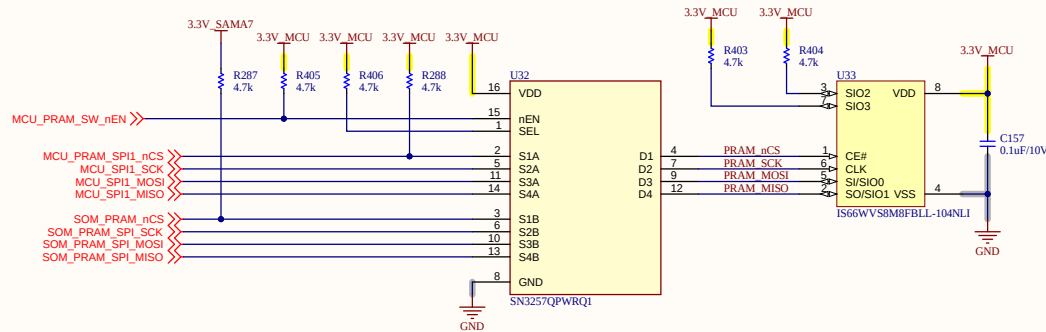
MRAM 4Mbit



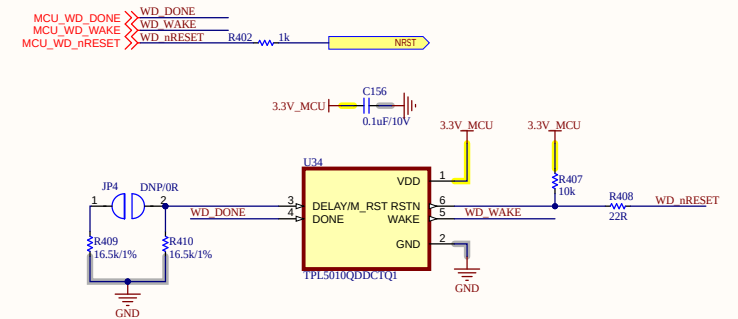
RTC



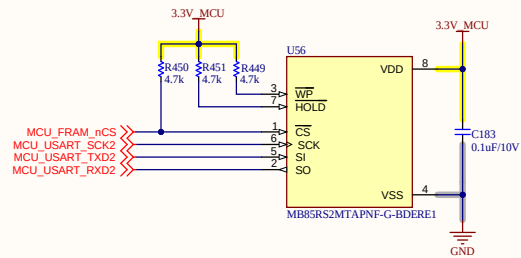
PRAM



WATCH DOG

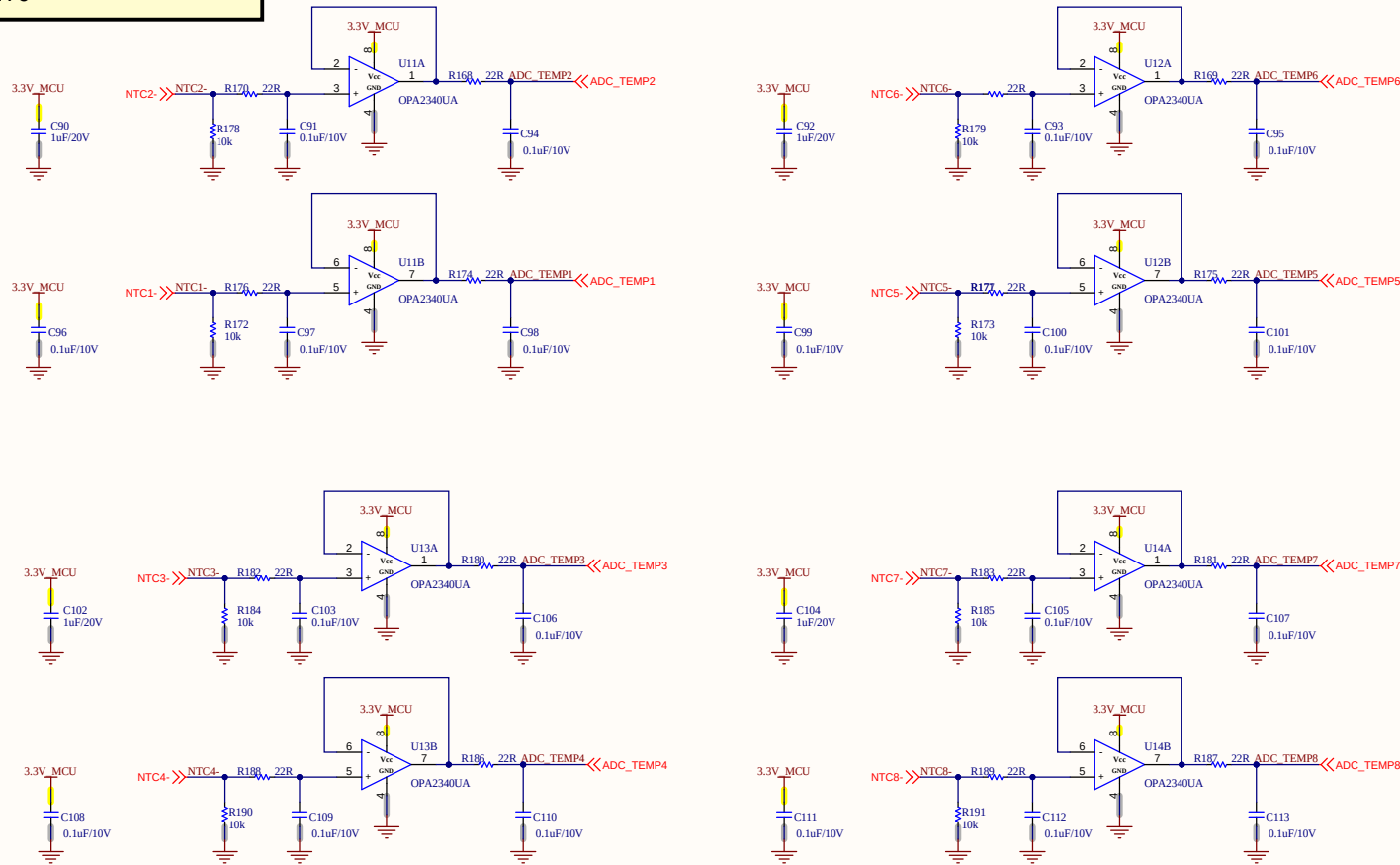


FRAM

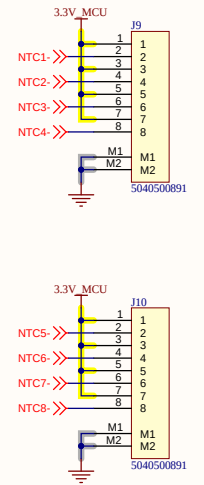


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 13_MRAM_RTC_WDT.Sch	Sheet 22 of 33
Checked by: BAO BUI	Document No: 22	Size: A3
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026

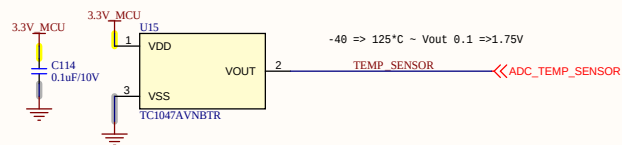
NTC



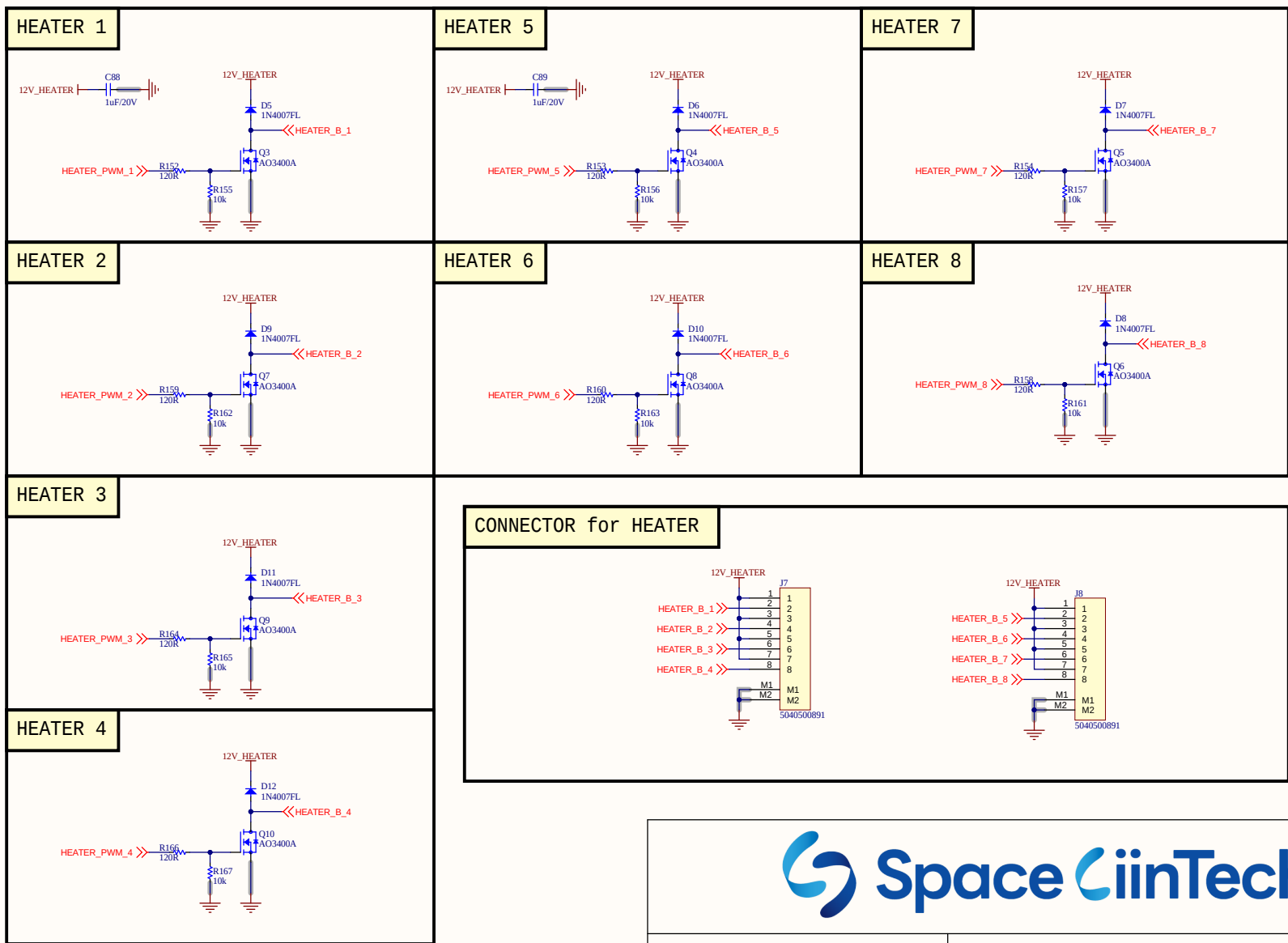
CONNECTOR FOR NTC



TEMPERATURE SENSOR

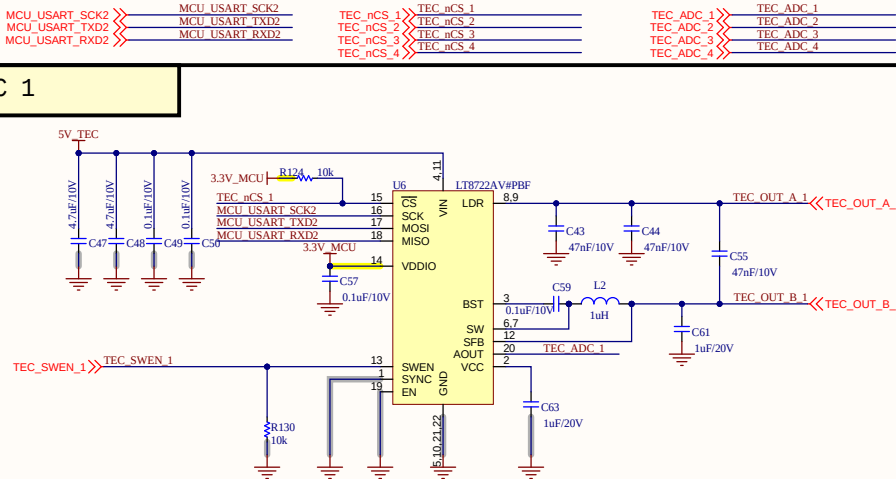


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 14_NTC_TEMPSENSOR	Sheet No: 23 of 33
Checked by: BAO BUI	Document No: 23	Size: A3
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026

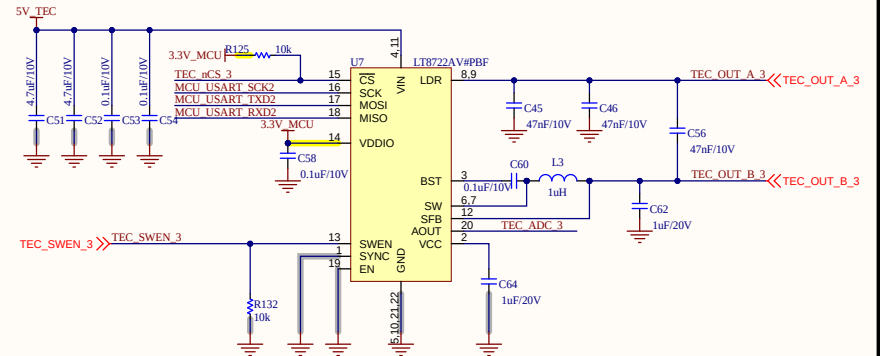


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 15_HEATER.SchDoc	Sheet 24 of 33
Checked by: BAO BUI	Document No: 24	Size: A3
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026

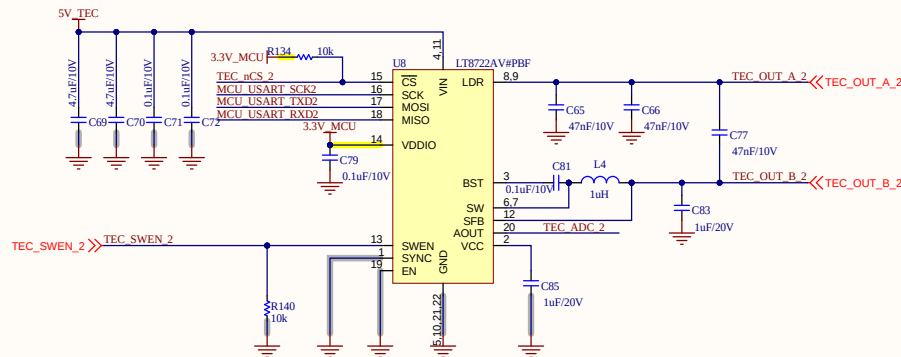
TEC 1



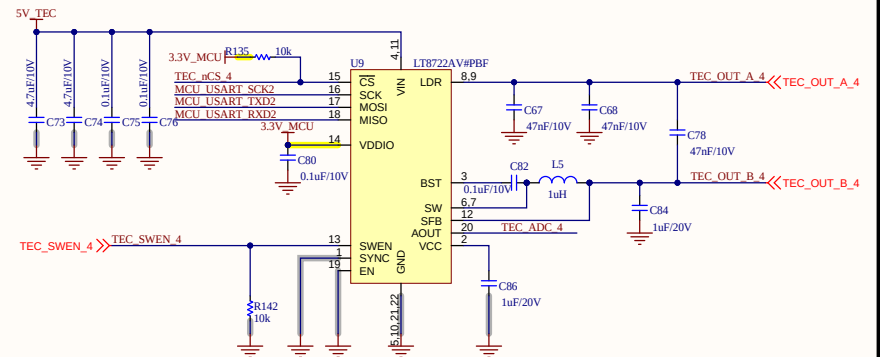
TEC 3



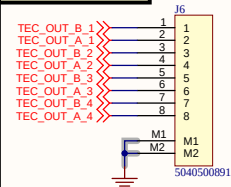
TEC 2



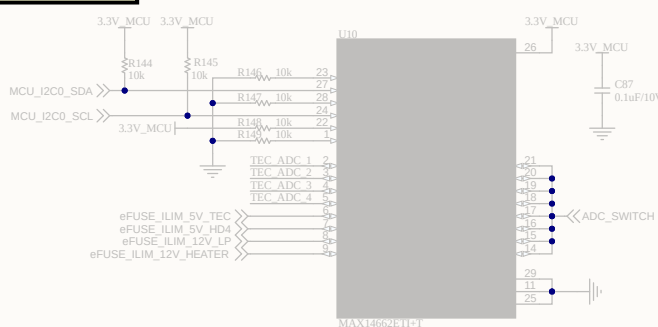
TEC 4



CONN TEC

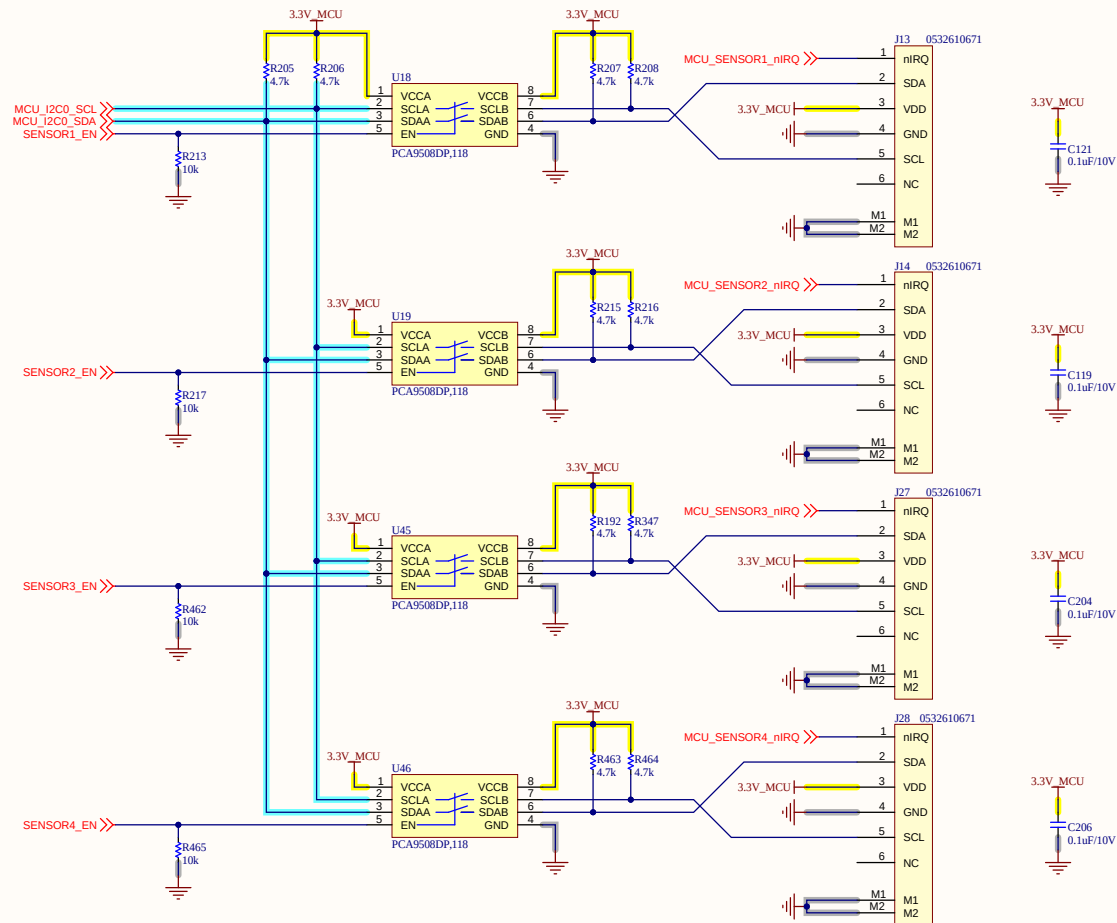


SWITCH

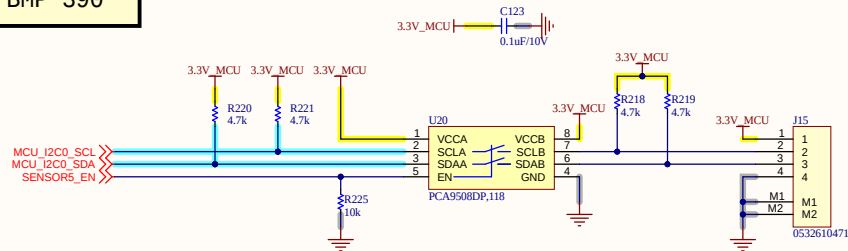


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 16_TEC.SchDoc	Sheet 25 of 33
Checked by: BAO BUI	Document No: 25	Size: A3
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026

I2C

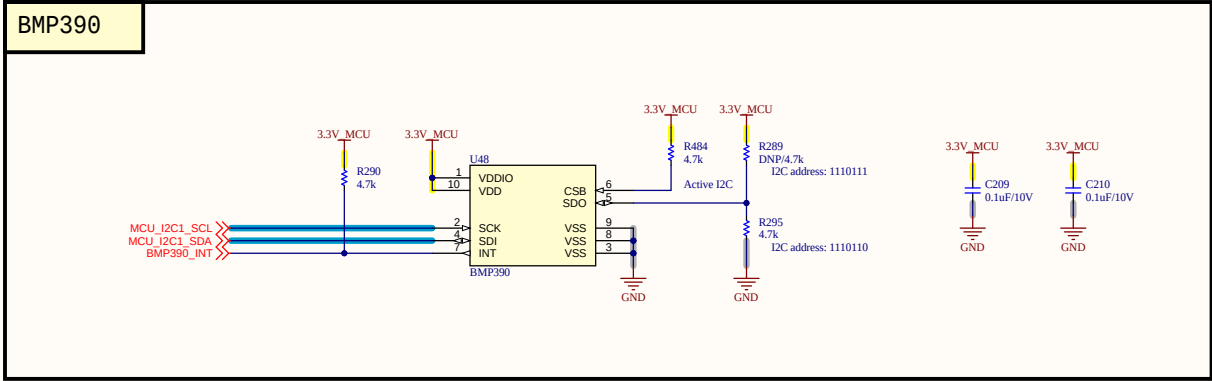


I2C BMP 390

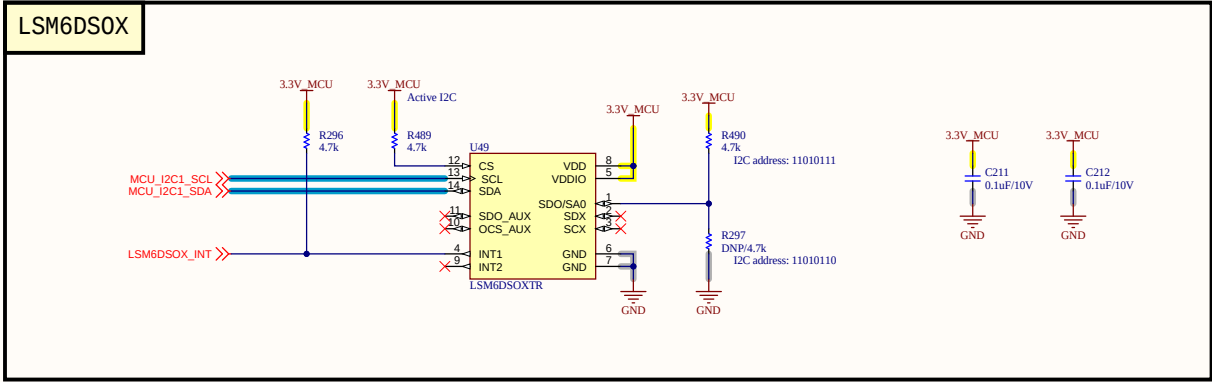


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER		
Design by: SANG HUYNH	Sheet title: 17_FLOW_Sensor.SchDoc Sheet 26 of 33		
Checked by: BAO BUI	Document No: 26	Size: A3	
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026	

BMP390



LSM6DS0X



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 17_I2C_Sensor_OnBoard.SchDoc	Sheet 27 of 33
---	----------------

Checked by:	BAO BUI
-------------	---------

Document No:	27
--------------	----

Size:	A3
-------	----

Approved by: **VU PHAM**

Revision: 1.0.0

Date: 1/23/2026

HIGH DRIVER 4

Diagram 1 (Left):

- Connector: J11
- Pin 1: HD4 CH31-
- Pin 2: HD4 CH31+
- Pin 3: HD4 CH32+
- Pin 4: HD4 CH32-
- Pin 5: HD4 CH41-
- Pin 6: HD4 CH41+
- Pin 7: HD4 CH42+
- Pin 8: HD4 CH42-
- Pin 9: M1
- Pin 10: M2
- Part Number: 5040500891

Diagram 2 (Right):

- Connector: J12
- Pin 1: HD4 CH22-
- Pin 2: HD4 CH22+
- Pin 3: HD4 CH21+
- Pin 4: HD4 CH21-
- Pin 5: HD4 CH12-
- Pin 6: HD4 CH12+
- Pin 7: HD4 CH11+
- Pin 8: HD4 CH11-
- Pin 9: M1
- Pin 10: M2
- Part Number: 5040500891

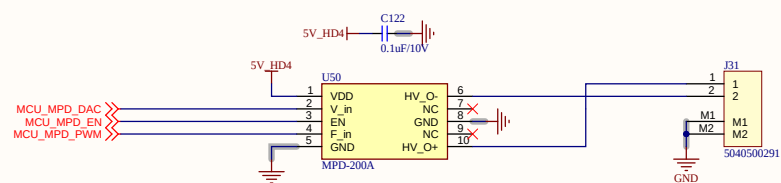


Sub-system: SATELLITE_EXP_CONTROLLER

Sheet 28 of 33

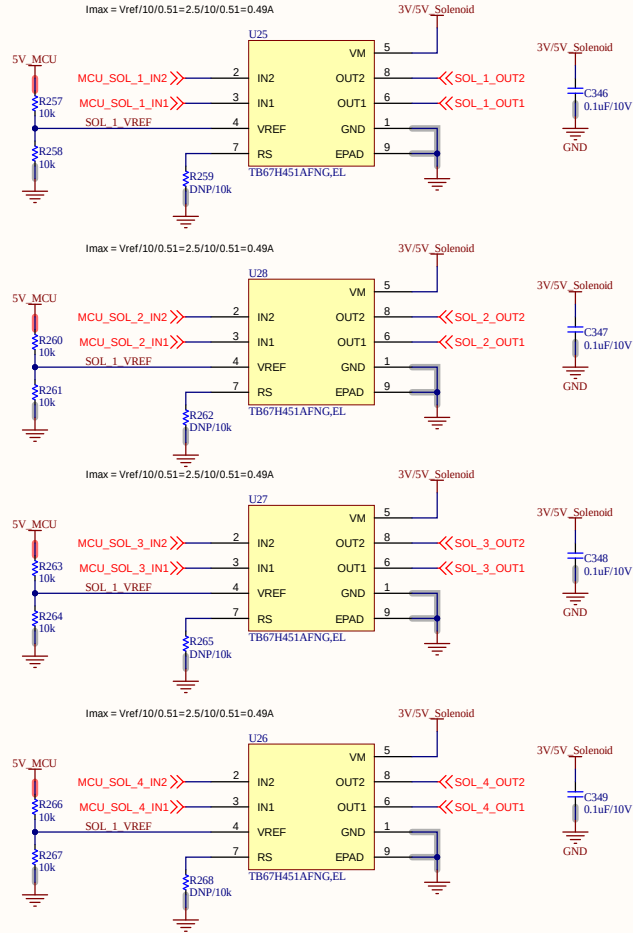
Size: A3

Date: 1/23/2026

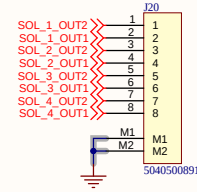


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER		
Design by: SANG HUYNH	Sheet title: 18_HD4_PUMP_2.SchDoc	Sheet 29	of 33
Checked by: BAO BUI	Document No: 29	Size: A3	
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026	

SOLENOID



CONNECTOR SOLENOID



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 19_SOLEDNOID_1.SchDocSheet 30 of 33

Checked by: BAO BUI

Document No: 30

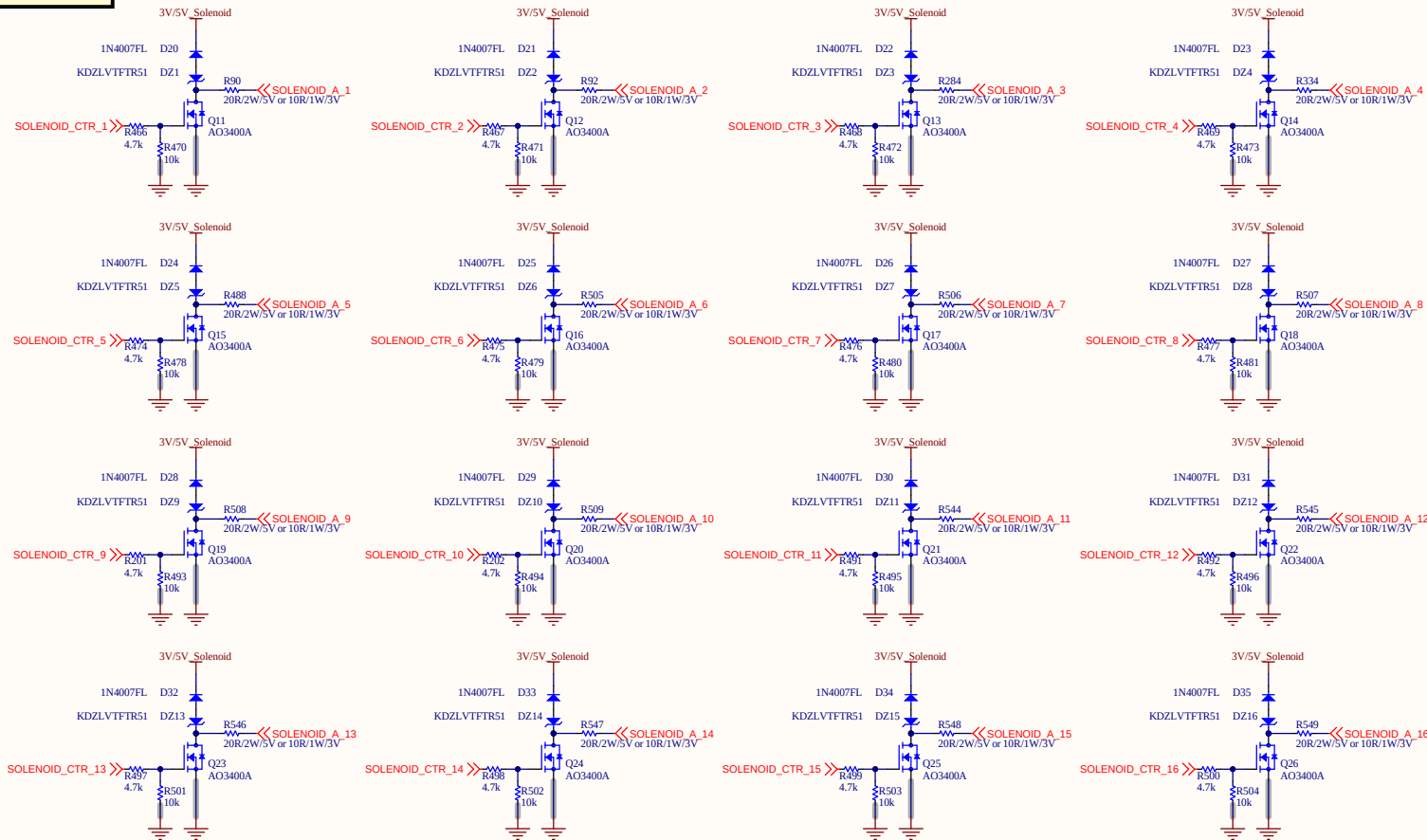
Size: A3

Approved by: VU PHAM

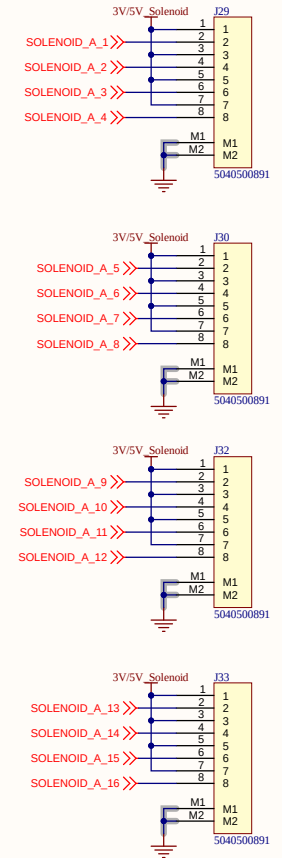
Revision: 1.0.0

Date: 1/23/2026

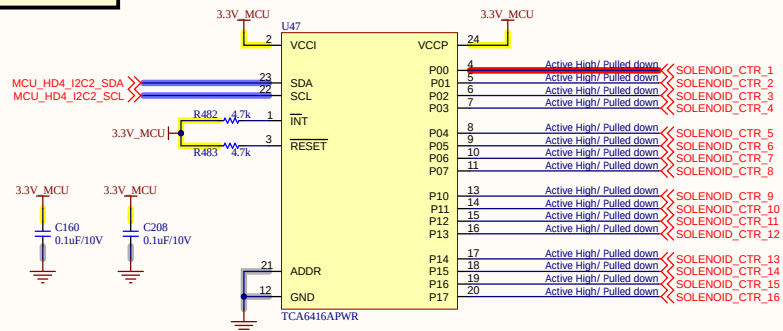
SOLENOID



CONN SOLENOID



I/O EXPANDED



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 19_SOLEDNOID_2.SchDocSheet 31 of 33

Checked by: BAO BUI

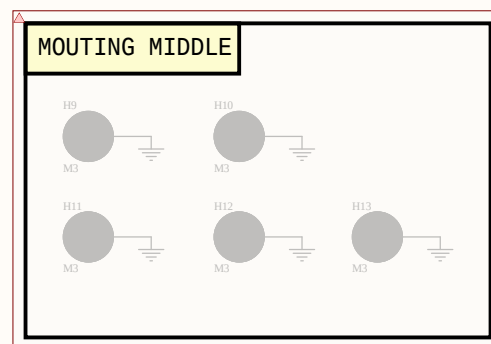
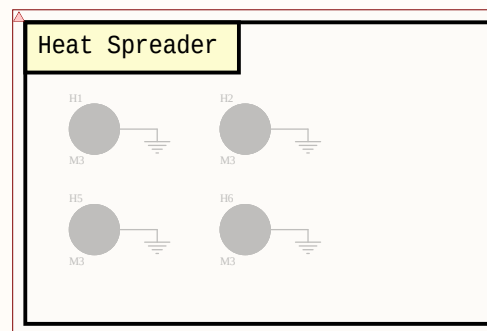
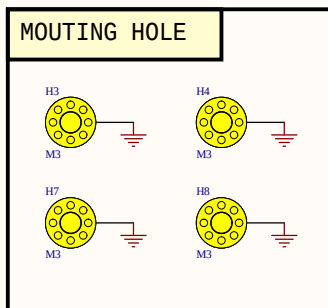
Document No: 31

Size: A3

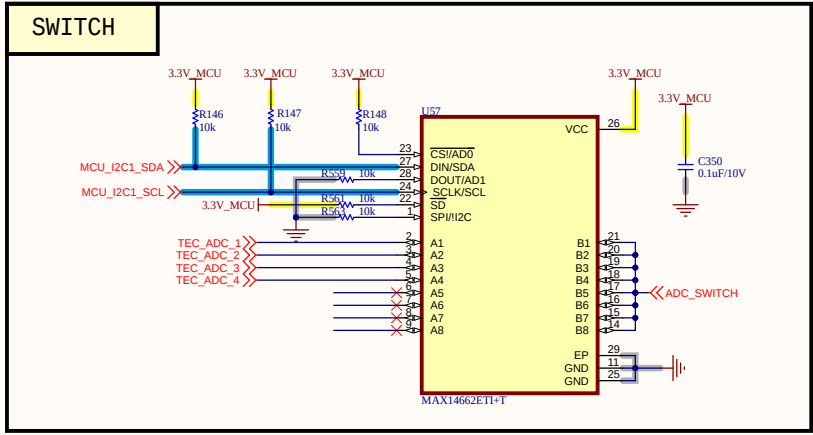
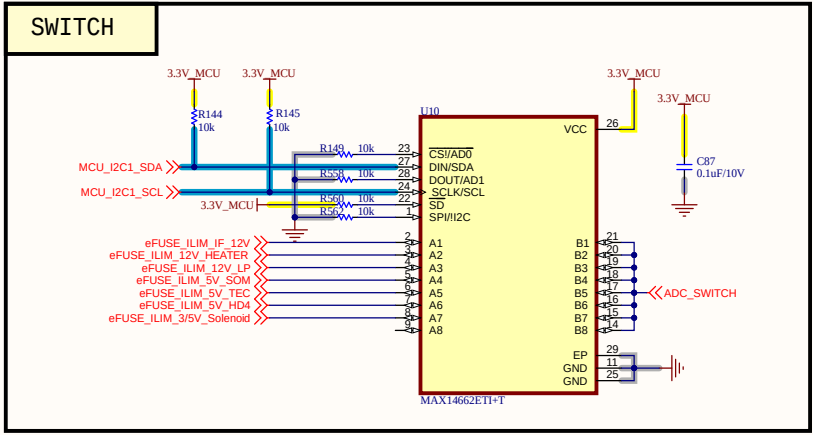
Approved by: VU PHAM

Revision: 1.0.0

Date: 1/23/2026



Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: MOUTING.SchDoc	Sheet 32 of 33	
Checked by: BAO BUI	Document No: 32	Size: A3	
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026	



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: ANA_SWITCH.SchDoc	Sheet 33 of 33
Checked by: BAO BUI	Document No: 33	Size: A3
Approved by: VU PHAM	Revision: 1.0.0	Date: 1/23/2026